

VECTOR DIFFERENTIATION

MODULE 4 CONTINUATION

Divergence and Curl of a vector function

If f be any given differentiable vector function then divergence of f denoted by $\nabla \cdot \bar{f}$ or $\text{div } f$ is defined as

$$\begin{aligned}\text{div } f &= \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \cdot f \\ &= \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \cdot (f_1 i + f_2 j + f_3 k) \\ &= \frac{\partial(f_1)}{\partial x} + \frac{\partial(f_2)}{\partial y} + \frac{\partial(f_3)}{\partial z}\end{aligned}$$

$$\nabla \cdot f = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z} + \frac{\partial f_3}{\partial z}$$

Solenoidal Vector

A vector function f is said to be solenoidal if $\text{div } \bar{f} = 0$

Curl f

If f be any given differentiable vector function then curl or notation of $\bar{f}$ written as $\nabla \times f$ or curl $\bar{f}$ is defined as

$$\text{curl } \bar{f} = \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \times \bar{f}$$

When $\vec{f} = f_1 \hat{i} + f_2 \hat{j} + f_3 \hat{k}$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f_1 & f_2 & f_3 \end{vmatrix}$$

$$= \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z} \right) \hat{i} + \left(\frac{\partial f_1}{\partial z} - \frac{\partial f_3}{\partial x} \right) \hat{j} + \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right) \hat{k}$$

$$\nabla \times \vec{f} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f_1 & f_2 & f_3 \end{vmatrix}$$

Irrrotational vector

A vector function f is said to be irrotational if $\text{curl } \vec{f} = 0$

Conservative field

A vector function f is said to form a conservative field if it is irrotational. i.e. $\text{curl } f = 0$

In this case we can find a scalar function $\phi(x, y, z)$ such that $\vec{f} = \nabla \phi$
This function ϕ is called scalar potential of $\vec{f}$

Problems

- 1) Compute the divergence and curl of the vector function $\vec{F} = xyz\mathbf{i} + 3x^2y\mathbf{j} + (xz^2 - y^2z)\mathbf{k}$ at the point $(1, 2, -1)$

$$\vec{F} = xyz\mathbf{i} + 3x^2y\mathbf{j} + (xz^2 - y^2z)\mathbf{k}$$

$$\text{div } \vec{F} = \nabla \cdot \vec{F} = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z}$$

$$= \frac{\partial}{\partial x}(xyz) + \frac{\partial}{\partial y}(3x^2y) + \frac{\partial}{\partial z}(xz^2 - y^2z)$$

$$= yz + 3x^2 + 2xz - y^2$$

$$\begin{aligned} \text{div } \vec{F} \Big|_{(1, 2, -1)} &= 2(-1) + 3(1)^2 + 2(1)(-1) - (2)^2 \\ &= -2 + 3 - 2 - 4 = \underline{\underline{-5}} \end{aligned}$$

$$\text{curl } \vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xyz & 3x^2y & xz^2 - y^2z \end{vmatrix}$$

$$\begin{aligned} &= \left[\frac{\partial}{\partial y}(xz^2 - y^2z) - \frac{\partial}{\partial z}(3x^2y) \right] \mathbf{i} - \frac{\partial}{\partial y} + \left[\frac{\partial}{\partial z}(xyz) - \frac{\partial}{\partial x}(xz^2 - y^2z) \right] \mathbf{j} \\ &\quad + \left[\frac{\partial}{\partial x}(3x^2y) - \frac{\partial}{\partial y}(xyz) \right] \mathbf{k} \end{aligned}$$

$$= -2yz\mathbf{i} + (xy - z^2)\mathbf{j} + (6xy - xz)\mathbf{k}$$

$$\text{curl } \vec{F} \Big|_{(1, 2, -1)} = \underline{\underline{4\mathbf{i} + \mathbf{j} + 13\mathbf{k}}}$$

2) Prove that (i) $\text{div } \vec{r} = 3$ ii) $\text{curl } \vec{r} = 0$

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\begin{aligned}\text{div } \vec{r} &= \nabla \cdot \vec{r} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot (x\hat{i} + y\hat{j} + z\hat{k}) \\ &= \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z) \\ &= 1 + 1 + 1 = \underline{\underline{3}}\end{aligned}$$

$$\begin{aligned}\text{curl } \vec{r} &= \nabla \times \vec{r} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & z \end{vmatrix} \\ &= \hat{i} \left[\frac{\partial}{\partial y}(z) - \frac{\partial}{\partial z}(y) \right] + \hat{j} \left[\frac{\partial}{\partial z}(x) - \frac{\partial}{\partial x}(z) \right] \\ &\quad + \hat{k} \left[\frac{\partial}{\partial x}(y) - \frac{\partial}{\partial y}(x) \right] = 0\end{aligned}$$

3) Determine the constant a so that the vector $\vec{f} = (x+3y)\hat{i} + (y-2z)\hat{j} + (x+az)\hat{k}$ is solenoidal

$$\vec{f} \text{ solenoidal} \Rightarrow \text{div } \vec{f} = 0$$

$$\begin{aligned}\text{div } \vec{f} &= \frac{\partial}{\partial x}[x+3y] + \frac{\partial}{\partial y}[y-2z] + \frac{\partial}{\partial z}[x+az] \\ &= 1 + 1 + a = 2 + a\end{aligned}$$

$$\begin{aligned}\text{div } \vec{f} = 0 &\Rightarrow 2 + a = 0 \\ &\Rightarrow \underline{\underline{a = -2}}\end{aligned}$$

4) Find the value of a if

$$\vec{A} = (axy - z^2)\mathbf{i} + (x^2 + 2yz)\mathbf{j} + (y^2 - axz)\mathbf{k}$$

is irrotational

$$\text{if irrotational} \Rightarrow \text{curl } \vec{f} = 0$$

$$\text{curl } \vec{f} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ axy - z^2 & x^2 + 2yz & y^2 - axz \end{vmatrix}$$

$$= \mathbf{i} \left[\frac{\partial}{\partial y} (y^2 - axz) - \frac{\partial}{\partial z} (x^2 + 2yz) \right] +$$

$$\mathbf{j} \left[\frac{\partial}{\partial z} (axy - z^2) - \frac{\partial}{\partial x} (y^2 - axz) \right] + \mathbf{k} \left[\frac{\partial}{\partial x} (x^2 + 2yz) - \frac{\partial}{\partial y} (axy - z^2) \right]$$

$$= \mathbf{i} [2y - 2y] + \mathbf{j} [-2z + az] + \mathbf{k} [2x - ax]$$

$$= \mathbf{j} (2z + az) + \mathbf{k} (2x - ax)$$

$$\text{curl } \vec{f} = 0 \Rightarrow az - 2z = 0 \quad -ax + 2x = 0$$

$$\Rightarrow a = 2$$

5) Show that the vector functional

$$\vec{F} = (6xy + z^3)\mathbf{i} + (3x^2 - z)\mathbf{j} + (3xz^2 - y)\mathbf{k}$$

forms a conservative force field and find its scalar potential.

$$\vec{F} = (6xy + z^3)\hat{i} + (3x^2 - z)\hat{j} + (3xz^2 - y)\hat{k} \quad \text{--- (1)}$$

$$\text{curl } \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 6xy + z^3 & 3x^2 - z & 3xz^2 - y \end{vmatrix}$$

$$= \hat{i} \left[\frac{\partial}{\partial y} (3xz^2 - y) - \frac{\partial}{\partial z} (3x^2 - z) \right] + \hat{j} \left[\frac{\partial}{\partial z} (6xy + z^3) - \frac{\partial}{\partial x} (3xz^2 - y) \right] + \hat{k} \left[\frac{\partial}{\partial x} (3x^2 - z) - \frac{\partial}{\partial y} (6xy + z^3) \right]$$

$$= \hat{i}(-1+1) + \hat{j}(3z^2-3z^2) + \hat{k}(6x-6x) = 0$$

Hence $\vec{F}$ is irrotational

$\therefore \vec{F}$ forms conservative force field.

We have to find a function ϕ such that $\vec{F} = \nabla\phi$ --- (2)

$$\nabla\phi = \hat{i} \frac{\partial\phi}{\partial x} + \hat{j} \frac{\partial\phi}{\partial y} + \hat{k} \frac{\partial\phi}{\partial z} \quad \text{--- (3)}$$

from (1) (2) & (3)

$$\frac{\partial\phi}{\partial x} = 6xy + z^3 \quad \frac{\partial\phi}{\partial y} = 3x^2 - z \quad \frac{\partial\phi}{\partial z} = 3xz^2 - y$$

(4) (5) (6)

$$\Rightarrow \int \frac{\partial\phi}{\partial x} = \int (6xy + z^3) dx \Rightarrow \phi = 3x^2y + z^3x + f_1(y, z) \quad \text{--- I}$$

$$\Rightarrow \int \frac{\partial\phi}{\partial y} = \int (3x^2 - z) dy \Rightarrow \phi = 3x^2y - zy + f_2(x, z) \quad \text{--- II}$$

$$\Rightarrow \int \frac{\partial\phi}{\partial z} = \int (3xz^2 - y) dz \Rightarrow \phi = xz^3 - yz + f_3(x, y) \quad \text{--- III}$$

Ex.

from I, II & III

$$\phi = 3x^2y + xz^3 - yz + C$$

Another method to find scalar potential

$$\begin{aligned} \phi = \int \frac{\partial \phi}{\partial x} dx + \int (\text{terms of } \frac{\partial \phi}{\partial y} \text{ not containing } x) dy \\ + \int (\text{terms of } \frac{\partial \phi}{\partial z} \text{ containing neither } x \text{ nor } y) dz \end{aligned}$$

In the above problem

$$\frac{\partial \phi}{\partial x} = 6xy + z^3, \quad \frac{\partial \phi}{\partial y} = 3x^2 - z, \quad \frac{\partial \phi}{\partial z} = 3x^2z - y$$

$$\begin{aligned} \phi &= \int (6xy + z^3) dx + \int (-z) dy + \int 0 dz \\ &= \underline{\underline{3x^2y + z^3x - zy + C}} \end{aligned}$$



THANK YOU

STUDENTS